Project Data Analysis & Synthesis



ACE

CYBER R&D Program

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Project Title:<Contextualized Threat Intelligence Prioritization for CTI analysts>

- ***1. Research Objective & Hypothesis Traceability***
- *Research Objective:*
  - *The objective of this research is to evaluate whether contextualized threat intelligence prioritization can reduce alert fatigue and improve efficiency for Cyber Threat Intelligence (CTI) analysts.*
- *Restate the research question(s)*
  - *Can contextualized automated threat prioritization reduce CTI analysts' workload?*
  - *Can automated prioritization maintain prioritization accuracy while improving efficiency?*
  - *Does contextual prioritization improve threat identification speed?*
- *Identify tested hypotheses*
  - *Hypothesis 1: Automated prioritization reduces the time required to identify high-priority threats*
  - *Hypothesis 2: Contextual prioritization reduces alert fatigue and analyst workload*
  - *Hypothesis 3: Automated prioritization maintains the same identification accuracy*
- *Map hypotheses to:*
- *Data sources*
- *Metrics*
- *Experimental scenarios*
  -

Hypothesis	*Data Source*	*Metric*	*Scenario*
Hypothesis 1	*Synthetic Logs*	*Time to identify high-priority logs*	*TC-001 & TC-002*
Hypothesis 2	*Synthetic Logs*	*Alert volume analysis*	*TC-001 & TC-002*
Hypothesis 3	*Classification Output*	*Accuracy*	*TC-002 & TC-003*

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- *Purpose: Ensures analysis stays anchored to the research problem.*
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- ***2. Data Description & Provenance***

- *Data sources (simulated, collected, public, synthetic)*
    - *Synthetic log data generated using Linux logger commands*
    - *Filebeat log ingestion pipeline*
    - *Kibana visualization and filtering*
    - *Elasticsearch log data*
- *Data timeframe*
    - *Controlled experiment runs*
    - *Generated within 15-minute to 1-hour windows*
- *Data volume*
    - *TC-001: 1000 logs*
    - *TC-002: 1065 logs*
    - *TC-003: 2000 logs*
- *Data preprocessing steps*
    - *Weighted contextual scoring model using severity indicators, contextual network, and account-based risk factors*
    - *Runtime field prioritization*
    - *Lowercase normalization is applied to log messages to ensure consistent scoring and classification*
- *Assumptions*
    - *Synthetic logs replicate real-world CTI environments*
    - *Testbed environment reflects analyst alert volume*
- *Constraints*
    - *Synthetic dataset*
    - *Limited virtual machine resources*
- *Purpose: Establishes credibility and repeatability.*
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- ***3. Analytical Methods Used***
- *Statistical methods, models, or analytical frameworks*
    - *Comparative analysis (Baseline vs Automated)*
    - *Time-based performance*
    - *Analysis and comparison accuracy*
- *Tools and environments (e.g., Python, cloud lab, SIEM)*
    - *Ubuntu 24.04*
    - *Docker Container*
    - *Elasticsearch*
    - *Kibana*
    - *Filebeat*

    - *Oracle Virtualbox*
- *Justification for method selection*
    - *These tools simulate a realistic CTI analyst workflow and provide a measurable evaluation.*
- *Purpose: Demonstrates methodological rigor.*
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- ***4. Metrics & Evaluation Criteria***
- *Definition of each metric*
    - *Time to identify threats*
        - *Evaluate efficiency improvements*
    - *Prioritization Accuracy*
        - *Accurate prioritization of log alerts*
    - *Alert Volume*
        - *Number of logs processed*
- *Why metrics are appropriate for the hypothesis*
    - *These metrics provide the necessary data to validate assumptions and develop a testable hypothesis.*
- *Thresholds or expected outcomes*
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Metric	*Expected Outcome*
Time	*Reduced*
Accuracy	*Conserved*
Alert Fatigue (Alert Volume)	*Reduced*

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- *Purpose: Prevents arbitrary success claims.*
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- ***5. Results & Observations***
- *Key findings (tables, figures, summarized outputs)*
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Test Case	*Logs*	*Average Time*	*Accuracy*
TC-001	*1000*	*3144 sec (~52.4 min)*	*Manual (Baseline)*
TC-002	*1065*	*6.8 sec*	*100%*
TC-003	*2000*	*5.8 sec*	*100%*

    - *Observed log count for TC-002 was approximately 1065 due to background system logs and missed ingestions within the Kibana time window, reflecting realistic, uncontrollable datasets. These results demonstrated a reduction in threat identification time of over 99% when using automated prioritization compared to a manual analysis.*
- *Patterns, anomalies, and unexpected behaviors*
    - *Minor errors in the log distribution and total count were observed due to background system logs and ingestion timing within the testing window.*
- *Observations grounded in evidence*
    - *Automated prioritization significantly reduced analysis time*
    - *Manual analysis resulted in increased alert fatigue*
    - *Increased log volume had little impact on system performance*
- *Purpose: Separates raw results from interpretation.*
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- ***6. Synthesis & Interpretation***
- *What the results mean relative to the hypothesis*
    - *The results demonstrate that contextualized threat intelligence prioritization significantly improves efficiency while maintaining prioritization accuracy.*
- *Cross-comparison of findings*
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Metric	*Manual*	*Automated*
Speed	*Slow*	*Fast*
Accuracy	*Varies*	*Consistent*
Workload	*High*	*Low*

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- *Alignment (or misalignment) with prior literature*
    - *Follows ongoing research and testing of hypotheses*
- *Mission relevance for DoD environments*
    - *Reduces analyst workload*
    - *Improves system detection speed*
    - *Enhances operational awareness*
- *Purpose: This is the most important TRL 2-3 section.*
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- ***7. Limitations, Biases, and Threats to Validity***
- *Experimental limitations*
  - *Synthetic data*
  - *Smaller-scale testbed & dataset*
  - *Limited threat intelligence integration*
- *Data quality issues*
  - *Limited log diversity*
  - *Rule-based contextual scoring model (simulated MISP-style prioritization)*
  - *Synthetic log data*
  - *Log ingestion delays*
  - *Time window selection*
- *Environmental constraints*
  - *May fail to resemble real-world environments*
- *External validity concerns*
  - *Real-world environments may behave differently*
  - *Larger datasets are yet to be tested*
- *Purpose: Shows research maturity and honesty.*
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- ***8. Conclusions & TRL Advancement Justification***
- *Final conclusions supported by evidence*
  - *Contextualized threat intelligence prioritization significantly improves CTI analyst efficiency and reduces alert fatigue while maintaining prioritization accuracy.*
- *Why this research qualifies as TRL 2-3*
  - *Proof of concept is demonstrated*
  - *Measurable results are documented*
  - *Controlled lab experimentation completed*
- *What additional work is required for TRL-4+*
  - *Real-world datasets*
  - *Further MISP integration*
  - *Operational testing*
- *Purpose: Makes TRL readiness explicit.*
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- ***9. Recommendations for Applied Follow-On Work***
- *Tool improvements*
  - *Further MISP scoring integration*

    - *Expanded datasets*
    - *Behavioral alert analysis*
- *Operational testing scenarios*
    - *Larger datasets*
    - *Real-world logs*
    - *Integration into existing DoD workflows*
- *Purpose: Feeds future R&D or sponsors.*
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- ***10. Appendices (Optional but Encouraged)***
- *Appendix A: Log Generation Script*
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```
#!/bin/bash

echo "Generating HIGH priority logs..."
for i in {1..100}; do
  logger "Multiple failed login attempts detected from 10.0.0.$i"
  logger "Unauthorized blocked access attempt from 10.0.0.$i"
  logger "Blocked admin login attempt from 10.0.0.$i port 22"
done

echo "Generating MEDIUM priority logs..."
for i in {1..100}; do
  logger "Failed password detected from 192.168.1.$i"
  logger "Invalid user detected from 172.16.0.$i"
  logger "Failed login attempt from 192.168.1.$i"
done

echo "Generating LOW priority logs..."
for i in {1..100}; do
  logger "System operation completed successfully"
  logger "Accepted password from user on network 192.168.10.$i"
  logger "Routine background process completed successfully"
  logger "Accepted user login from 192.168.10.$i"
done

echo "Log Generation Complete."
```
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- *Appendix B: Runtime Field Script*
    -
```
String msg = params._source['message'];
```

```
int score = 0;

if (msg != null) {
    msg = msg.toLowerCase();

    // Severity Indicators
    if (msg.contains("blocked")) score += 60;
    if (msg.contains("multiple")) score += 50;
    if (msg.contains("denied")) score += 40;
    if (msg.contains("unauthorized")) score += 35;
    if (msg.contains("failed")) score += 30;
    if (msg.contains("invalid")) score += 20;



    // Account Verification
    if (msg.contains("admin")) score += 15;
    if (msg.contains("root")) score += 15;
    if (msg.contains("port 22")) score += 10;

    // Networks
    if (msg.contains("10.0.0")) score += 15;
    if (msg.contains("192.168")) score += 10;
    if (msg.contains("172.16")) score += 10;
    if (msg.contains("127.0.0")) score += 5;

    // Low Risk
    if (msg.contains("accepted")) score -= 10;
    if (msg.contains("success")) score -= 10;
    if (msg.contains("completed")) score -= 10;

    // Final Calculated Priority
    if (score >= 70) {
        emit("HIGH");
    } else if (score >= 30) {
        emit("MEDIUM");
    } else {
        emit("LOW");
    }
}
```

- *Appendix C: Experiment Results Table*

Test Case	*Logs*	*Average Time*	*Accuracy*
TC-001	*1000*	*3144 sec (~52.4 min)*	*Manual (Baseline)*
TC-002	*1065*	*6.8 sec*	*100%*
TC-003	*2000*	*5.8 sec*	*100%*

- *Appendix D: Testbed Architecture*

